

PROFESSIONAL SUMMARY

Graduate student in Applied Computing specializing in data engineering, analytics, and workflow optimization. Experienced in designing and managing end-to-end data workflows for data collection, processing, and reporting, with a focus on data quality, KPI tracking, and operational performance improvement. Skilled in Python, SQL, Azure, and Tableau to deliver actionable insights and scalable data solutions supporting business decision-making.

SKILLS

Programming: Python, SQL, Java
Data Engineering: ETL Pipelines, Data Modeling, Data Warehousing, Lakehouse Architecture (Bronze/Silver/Gold), Data Validation
Streaming & Big Data: Apache Kafka, Apache Spark, Real-Time Data Processing
Workflow Orchestration: Apache Airflow (DAGs, Scheduling, Monitoring)
Databases: MySQL, PostgreSQL, MongoDB, Azure SQL
Cloud Platforms: Microsoft Azure (Data Lake, Blob Storage, SQL Database), AWS (EC2, S3)
Machine Learning: KMeans (Segmentation), Isolation Forest, LOF, One-Class SVM (Anomaly Detection), Random Forest (Risk Scoring)
MLOps: Model Monitoring, Retraining Pipelines, Feature Engineering, Versioned Deployment
Visualization: Tableau, Power BI
Tools: Git/GitHub, Jupyter Notebook, VS Code
Concepts: Data Quality, KPI Monitoring, Fraud Detection, Risk Analytics, Scalable Data Systems, AI-driven Analytics

ACADEMIC PROJECTS

Real-Time Banking Data Engineering Platform (Streaming & Lakehouse Architecture) 2025 – 2026

- Designed and managed end-to-end data workflows for real-time transaction ingestion, processing, and analytics using Python and SQL.
- Built automated ETL pipelines to ensure reliable data collection, transformation, and validation across distributed systems.
- Defined and tracked key performance indicators (KPIs) including processing latency, data accuracy, and pipeline reliability.
- Developed analytics-ready datasets to support reporting, anomaly detection, and data-driven decision-making.
- Implemented data quality validation frameworks to ensure consistency, integrity, and governance of data assets.
- Optimized workflow processes, reducing manual reporting effort and improving overall system efficiency.

Limited Random Forest: Enhancing ML Performance and Efficiency (Research Project) 2023 – 2024

- Developed an enhanced version of Random Forest classifier improving prediction accuracy and computational efficiency.
- Designed a custom feature importance mechanism by averaging importance across individual trees for better representation.
- Implemented a modified splitting strategy prioritizing the most important feature at the root node, reducing model randomness.
- Achieved 97.96% accuracy and F1-score, outperforming standard Random Forest and other classification algorithms.
- Conducted comparative analysis with Logistic Regression, K-Nearest Neighbors, Decision Tree, and Naïve Bayes on a large diabetes dataset.
- Integrated the model with `scikit-learn` pipeline ensuring scalability and compatibility for real-world applications.

PROFESSIONAL EXPERIENCE

AI/ML Intern May 2023 – Apr 2024
Bhavi Technologies Pvt. Ltd., India

- Built and maintained data pipelines supporting data annotation, preprocessing, and model training workflows.
- Ensured data quality through validation and transformation, improving dataset reliability for machine learning tasks.
- Optimized analytical workflows by enhancing feature engineering processes, reducing execution time by 25%.
- Conducted data analysis to generate insights for improving model performance and operational efficiency.
- Collaborated with cross-functional teams to translate analytical outputs into actionable business insights.

Project Intern Jun 2022 – Aug 2022
Computer Centre, MSU Baroda, India

- Designed and executed SQL queries to manage relational databases and support structured data systems.
- Assisted in backend development using ASP.NET and MySQL with focus on workflow efficiency and data integrity.
- Supported testing, troubleshooting, and documentation to improve system performance and reliability.

EDUCATION

Master of Applied Computing, Wilfrid Laurier University Jan 2025 – Apr 2026
Bachelor of Technology, Computer Science & Engineering, MSU Baroda - CGPA: 9.08 / 10.0 Aug 2020 – Apr 2024